

# DSAA3073: Theories in Data Science

## Week 8 Validation Sheet · Margins, Surrogates, and Evidence Boundaries

**Coverage:** Week 8

**Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

### Part A: Multiple choice

**A1 [5 marks].** Let  $F_{\alpha(x)} = \sum_t \alpha_t h_{t(x)}$  with  $\alpha \neq 0$ . Which quantity is unchanged when  $\alpha$  is replaced by  $c\alpha$  for any  $c > 0$  and still records whether  $(x, y)$  is classified correctly?

- (a)  $yF_{\alpha(x)}$
- (b)  $|F_{\alpha(x)}|$
- (c)  $y \frac{F_{\alpha(x)}}{\|\alpha\|_1}$
- (d) The Euclidean distance from the original input  $x$  to a decision boundary

**A2 [5 marks].** Which statement about  $\hat{R}_{S,\theta} = m^{-1} \sum_i \mathbf{1}[\rho_i \leq \theta]$  and the Week 8 margin certificate is correct?

- (a)  $\hat{R}_{S,\theta}$  must decrease as  $\theta$  increases.
- (b) The displayed theorem fixes one positive  $\theta$  before its probability statement.
- (c)  $\hat{R}_{S,\theta}$  is an observed test-error curve.
- (d) Uniformity over normalized ensemble functions automatically permits choosing any  $\theta$  after seeing the sample.

**A3 [5 marks].** From  $\mathbf{1}[yF(x) \leq 0] \leq e^{-yF(x)}$ , which conclusion follows on a fixed training sample without any further theorem?

- (a) Empirical zero-one error is at most empirical exponential loss.
- (b) Population error equals empirical exponential loss.
- (c) Exponential-loss minimization is robust to every type of label noise.
- (d) Every optimizer of exponential loss follows AdaBoost's coefficient path.

**A4 [5 marks].** At the current score, suppose the directional derivative along coordinate  $h_k$  is  $C_{t(2\varepsilon_{t,k}-1)}$ , where  $C_t > 0$ . Which claim is licensed?

- (a) The coordinate with the smallest current weighted error has the most negative derivative.
- (b) The common factor  $C_t$  may be declared equal to one in the derivative identity.
- (c) Choosing the steepest coordinate proves the fitted model has the lowest test error.
- (d) Any coordinate schedule therefore reproduces AdaBoost's entire path.

## Part B: Definition in use

**B1 [20 marks].** A binary ensemble has coefficient vector  $\alpha = (2, -1, 1)$  and, at one input, weak-hypothesis-output vector  $h(x) = (1, -1, -1)$ . The label is  $y = -1$ .

Compute  $F_{\alpha(x)}$ ,  $\|\alpha\|_1$ , the normalized functional margin, and the unsigned  $L^\infty$  distance from  $h(x)$  to the hyperplane  $\{z : \alpha \cdot z = 0\}$ . Repeat the margin calculation after replacing  $\alpha$  by  $3\alpha$ . Then explain precisely why neither computed quantity is an input-space geometric distance and why the signed margin and unsigned output-space distance have different meanings on this example.

*Answer space*

## Part C: Basic skills

**C1 [7 marks].** Ten normalized training margins are

$$-0.10, 0.05, 0.20, 0.20, 0.35, 0.55, 0.60, 0.80, 0.90, 1.00.$$

At threshold  $\theta = 0.20$ , compute  $\hat{R}_{S,\theta}$ . If the two nonempirical terms in the fixed-threshold bound are 0.18 and 0.07, compute the resulting population-error upper bound.

*Answer space*

**C2 [8 marks].** A fitted score has empirical exponential loss 0.18. At one coordinate step,  $C_t = 0.24$  and the current weighted error is  $\varepsilon_{t,k} = 0.35$ . Give the empirical zero-one-error conclusion and compute the directional derivative  $C_{t(2\varepsilon_{t,k}-1)}$ . Interpret its sign.

*Answer space*

## Part D: Construction and diagnosis

D1 [17 marks]. On two inputs  $a, b$ , let

$$(h_1(a), h_1(b)) = (1, 1), \quad (h_2(a), h_2(b)) = (1, -1).$$

Consider the signed-coefficient normalized score  $G = \frac{2h_1 - h_2}{3}$ .

Use these two input values to prove that  $G$  is not in  $\text{conv}(\{h_1, h_2\})$  under the Week 8 convention of nonnegative weights whose sum is at most one. Then construct a valid convex-hull representation after making the dictionary sign-closed, and explain why division by an  $L^1$  coefficient norm did not by itself license the original convex-hull claim.

*Answer space*

**D2 [18 marks].** During several boosting rounds, one training example's normalized weight rises from 0.03 to 0.41 while its signed score remains negative. No label audit or held-out results are available.

Construct two distinct data-generating or weak-class explanations compatible with this same trace: one in which the recorded label is wrong and one in which it is correct. For each explanation, identify evidence that could distinguish it from the other. Finally state exactly what the weight trace establishes and what it cannot establish about label correctness and test performance.

*Answer space*

## Part E: Proof and derivation

**E1 [25 marks].** For each  $\theta \in \{0.20, 0.50\}$  fixed before sampling, a current theorem states that, with probability at least  $1 - \eta$ , simultaneously for every  $\bar{F}$  in the normalized ensemble class,

$$R(\bar{F}) \leq \hat{R}_{S,\theta}(\bar{F}) + \frac{0.04}{\theta} + \sqrt{\frac{\log\left(\frac{1}{\eta}\right)}{400}}.$$

After seeing one fitted ensemble's training-margin curve, an analyst obtains  $\hat{R}_{S,0.20} = 0.08$  and  $\hat{R}_{S,0.50} = 0.30$ . The analyst uses  $\eta = 0.10$  separately for both candidates, selects the smaller right-hand side, and reports a 90% confidence certificate.

Audit this report and construct a numerical certificate that legally permits selection after both empirical values are seen while retaining overall confidence at least 90%. State the probability event and order of choices.

**Available fact.**  $\Pr(\cup_j B_j) \leq \sum_j \Pr(B_j)$  for finitely many bad events. Use  $\log(10) \approx 2.303$  and  $\log(20) \approx 2.996$ .

*Answer space*

## Part F: Synthesis and limitations

**F1 [15 marks].** A report about one fitted boosting model contains four sentences:

1. Its empirical exponential loss is 0.06, so its empirical zero-one error is at most 0.06.
2. Any other optimizer reaching objective value 0.06 must have followed the same AdaBoost path and produced the same classifier.
3. A valid fixed-threshold margin bound has right-hand side 0.24, so the model's observed test error equals 0.24.
4. An untouched held-out set gives error 0.11, so every future sample and every boosting problem must have error 0.11.

For each sentence, identify the relevant evidence type: objective value, AdaBoost update path, margin certificate, or held-out performance. Then write the strongest conclusion that the evidence actually supports. State any missing conditions needed for the stronger claims.

*Answer space*